2 A copper cooking pan contains water. Fig. 2.1 shows the pan on a hotplate of a cooker.

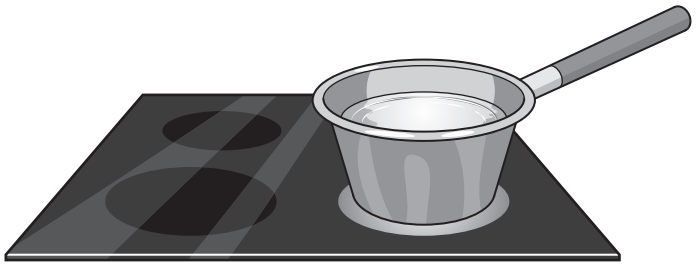

Fig. 2.1

Copper is a metal.

(a) Thermal energy is conducted through all solids by lattice vibrations.

Describe **one** other way in which thermal energy is conducted through the copper.

.....

.....

.....

..... [3]

(b) The outside surface of the cooking pan is kept clean by regular polishing.

Explain **one** other advantage of keeping the surface of the pan shiny.

.....

.....

..... [2]

(c) The thermal energy passes into the water through the base of the pan.

Identify the main method by which thermal energy is transferred throughout the water.

..... [1]

[Total: 6]

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